

Lab Report Instructions

A lab report is a structured way of communicating scientific research. Conducting research and then writing a lab report is part of learning to be a scientist and provides you with experience in collecting, organising, analysing and interpreting data and then writing this up in a scientific style. Throughout your undergraduate degree you will develop these skills and write more extended and increasingly original research reports.

Due Date and weighting

- Your lab report is **due Friday Week 10, 5 pm** via the Turnitin submission link in the Assessments section of Moodle.
- Your lab report assessment is worth 30% in total. The lab report is worth 25% and the lab project notes are worth 5%.
- You only get **ONE** submission. Please ensure you check the name of your file before upload so that you upload the correct file. You should also check you receive a receipt of submission, and you can click on your submission to see that it is displaying correctly for your marker.

Instructions

- As an individual you are required to write a 2000-word lab report on the BABS1202 laboratory project.
- During your laboratory classes, you will isolate a bacteria that is capable of degrading starch at high temperatures. Each week a portion of your laboratory class work will be dedicated to this project.
- You will also be assessed on your record keeping of the project. You will need to keep your records in a separate page of your Class Notebook each week that records your results from the laboratory work.
- Any changes to methods provided in the Laboratory Notebook and any technical errors that occur should be discussed in the report.
- The project spans across the whole term so **you should be writing your lab report throughout the term**
- Once you have completed the project, finalise the report and submit it via the Turnitin link on Moodle.
- The report should reference TWO scientific articles. You can use more articles.

Criteria

Your lab report will be assessed on the following criteria, which are included in the marking rubric found on the Assessments Hub of Moodle:

1. **Title.** The title should be descriptive of the project and clearly outline the topic of the lab report.
2. **Introduction.** The introduction of a lab report states the objectives/ aims of the experiment and provides the reader with relevant background information. Do not copy the Class Notebook, but show your understanding of the project and write this in your own words.

3. **Materials and methods.** Your method section should clearly outline the experimental procedures in paragraph form. These should include enough detail so that the reader could replicate the experiment.
4. **Results.** Key results are presented logically and clearly in both the appropriate illustrative manner (figures/ graphs/ tables) and also written text. All figures, graphs, and tables are presented correctly, numbered with appropriate captions and explicitly mentioned in the text.
5. **Discussion.** The discussion is a very important part of your lab report, because here you demonstrate that you have not merely completed the project, but that you also understand its wider implications and significance. Here the experimental results are interpreted and discussed using relevant terms and/ or theories. Any limitations or sources of errors mentioned. The discussion finishes with a conclusion which clearly links the aims to the interpretation of the results presented in the report and discusses viable applications or real-life significance of the results and possible future directions.
6. **Readability.** Writing is clear, and succinct with proofreading performed. The material is well organised and logically structured. Word count is adhered to.
7. **References.** All in-text references should be present in the reference list and formatted correctly, using Harvard style. For some help on how to reference in Harvard style please see <https://www.student.unsw.edu.au/harvard-referencing>, and <https://www.student.unsw.edu.au/how-do-i-cite>.
8. **Lab project notes.** Your lab project notes should be recorded in the “Lab Project Notes” section of your Class Notebook. Ensure they are organised using the appropriate headings and subheadings from the provided template, which can be found in the “Introduction to Lab Project Notes” tab of the same section. Notes relevant to the BABS1202 Lab Project parts of every lab should be recorded accurately using the template headings. There is no need to submit your lab notes to Turnitin; your demonstrator will review and mark your lab notes in your Class Notebook at the end of the term.

References

You must reference at least TWO scientific articles in your lab report using Harvard style. Below are some references you can use, or you may find some relevant articles referenced *within* these articles that you can use. You are also welcome to find and use other scientific articles if you prefer.

- Nimisha, P., Moksha, S., & Gangwane, A. K. (2019). Amylase activity of starch degrading bacteria isolated from soil. *Journal homepage: http://www.ijcmas.com*, 8(04), 2019.
- Azzopardi, E., Lloyd, C., Teixeira, S. R., Conlan, R. S., & Whitaker, I. S. (2016). Clinical applications of amylase: Novel perspectives. *Surgery*, 160(1), 26-37.
- Souza, P. M. D., & Magalhães, P. D. O. (2010). Application of microbial α -amylase in industry-A review. *Brazilian journal of microbiology*, 41, 850-861.
- Naidu, M. A., & Saranraj, P. (2013). Bacterial amylase: a review. *Int J Pharm Biol Arch*, 4(2), 274-87.

Word count

The word count is 2000. This excludes the reference section but includes in-text citations. You must perform a word count and indicate the count at the beginning of your lab report. Please note that any words beyond the 2000-word limit will not be read or marked by your assessor.

Plagiarism

Please revise the information on academic integrity and plagiarism here before commencing your literature review: <https://student.unsw.edu.au/plagiarism>

Tip: Plagiarism does not simply mean you must reference your sources. It means that your assignment must be *in your own words*. For help on using your own words and style in academic writing see <https://www.student.unsw.edu.au/intellectual-insecurity>. Please contact Academic Skills if you require advice on using your own words and developing your own style: <https://www.student.unsw.edu.au/skills>

Generative AI

PLANNING ASSISTANCE

As this assessment task involves some planning or creative processes, you are permitted to use software to generate initial ideas. However, you must develop or edit those ideas to such a significant extent that what is submitted is your own work, i.e. only occasional AI generated words or phrases may form part of your final submission. It is a good idea to keep copies of the initial prompts to show your lecturer if there is any uncertainty about the originality of your work. To cite: OpenAI (Year Accessed). ChatGPT. OpenAI. <https://openai.com/models/chatgpt/>

You can use generative AI software in this assessment for assistance with summarising your journal articles from the more technical language used in scientific literature. Any output of generative software used must be attributed with full referencing.

If the outputs of generative AI such as ChatGPT form part of your submission and is not appropriately attributed, it will be regarded as serious academic misconduct and subject to the standard penalties, which may include OOF, suspension and exclusion.

Important: You should ensure that you have moderated and critically evaluated the outputs from generative AI tools such as ChatGPT before submission, and keep a record of this for if you are required to show the process you used. Turnitin includes an AI detector and you will need to demonstrate that any work submitted is your own writing.

Learning Outcomes

1. Demonstrate a knowledge of the skills, techniques, experimental design and safe work practices of a bioscience laboratory, including fundamental microbiology, recombinant DNA and bioinformatics techniques.
2. Effectively communicate experimental results and findings from scientific literature using appropriate referencing to different audiences.
3. Demonstrate quantitative skills applicable to the biosciences, including serial dilutions, metric conversions, biochemical calculations data collection and organisation, graphing, and experimental design.

Need more help?

Ask questions on the forum, in your laboratory class and in the course Q&A sessions. For extra help on what to put in each section and how to write a simple lab report see: <https://www.student.unsw.edu.au/simple-lab-report>